

Estimate vs Bill Reconciliation Tool – Project Document

Purpose

The tool checks whether each patient's final bill matches the amount given in their estimate.

It reads Estimate Performa PDFs, saves the details in Excel, then reads the Billing PDFs and adds the billed amount against the same patient. Each patient gets a status, and a dashboard shows the totals.

Inputs

Document	What it is	Used for
Estimate Performa PDF	The estimate given to the patient before treatment	Creating the patient record in Excel with the estimate amount
Billing PDF (IP Bill)	The final bill raised for the patient	Adding the billed amount to the matching patient

Estimate PDFs and Billing PDFs are kept in two separate folders. The user selects each folder separately.

```
flowchart LR
    A[Select Estimate folder] --> B[Read estimate PDFs]
    B --> C[Add new patients<br/>to Excel]
    D[Select Billing folder] --> E[Read name + MRD<br/>+ billed amount]
    E --> F[Match with Excel]
    F --> G[Add billed amount<br/>+ set status]
    G --> H[Show results]
```

The estimate folder is read first, so patients are in Excel before their bills are matched.

Step 1: Read Estimate Performa PDFs

The tool reads every Estimate Performa PDF and saves one row per patient in Excel.

1. The user selects the Estimate folder.
2. The tool reads each PDF and picks out details such as patient name, MRD number, estimate date, department or procedure, doctor, and estimate amount.
3. Each new patient is added as a new row in Excel, with these details in their columns.

The exact fields will be confirmed from the sample Estimate Performa PDF.

Step 2: Read Billing PDFs and match patients

The tool reads every Billing PDF and adds the billed amount to the matching patient in Excel.

1. The user selects the Billing folder.
2. The tool reads the patient name, MRD number and billed amount from each PDF.
3. It looks for the same patient name and MRD number in Excel.
4. If the patient is found, the billed amount is added in the **Billed Amount** column for that patient.
5. The billed amount is then compared with the estimate amount and the status is set (Step 3).

Step 3: Set the status

After the billed amount is added, the tool compares it with the estimate amount and sets the status.

Condition	Status
Billed amount is the same as estimate amount	Closed
Billed amount is less than estimate amount	Less Amount
No bill found yet for the patient	Pending

Open question: what status should be used if the billed amount is more than the estimate?

Dashboard

The dashboard lets the user pick a folder and see the results once reading is done.

- **Select Estimate Folder** button: reads the estimate PDFs and adds new patients to Excel.
- **Select Billing Folder** button: reads the billing PDFs, adds billed amounts and sets the status.

After each billing run, the screen shows the results of that run:

- **Closed in this run:** list of patients whose status became Closed, with name, MRD number, estimate amount and billed amount.
- **Less amount in this run:** list of patients billed less than the estimate, with name, MRD number, estimate amount, billed amount and difference.

The screen also shows overall counts from the whole Excel file:

Item	Count of
Total patients	All patients in Excel
Closed	Patients with status Closed
Less Amount	Patients billed less than the estimate
Pending	Patients with no bill yet

Excel output

The Excel file keeps one row per patient with these columns. Final columns depend on what the estimate PDF holds.

Column	Filled from
Patient Name	Estimate PDF
MRD Number	Estimate PDF
Estimate Date	Estimate PDF
Department / Procedure	Estimate PDF
Estimate Amount	Estimate PDF
Billed Amount	Billing PDF

Column	Filled from
Difference	Estimate Amount minus Billed Amount
Status	Closed, Less Amount or Pending